

SPRING BOOT

CHAPTER – I

INTRODUCTION

SPRING:

- Spring is a Java-based framework used to develop enterprise applications, especially backend applications and web applications.

FRAMEWORK:

- A framework provides an environment and tools to build applications.

SPRING INTERNALS:

- Spring contains a lot of classes and interfaces
- These are provided as JAR files

JAR FILE:

- JAR = Java Archive File
- It contains:
 - Classes
 - Interfaces
 - Configuration needed to run Java applications

SPRING FEATURES AND CONCEPTS:

- **POJO:**
 - POJO class file contains:
 - ✓ Private variables
 - ✓ Getters and setters
 - All functions are written inside POJO files
 - Spring creates and manages POJO files using Spring configuration

➤ **Dependency injection:**

- Spring provides Dependency Injection (DI)
- Objects are created and dependencies are injected automatically by Spring

➤ **REST API:**

- Spring supports building RESTful APIs
- Used to transfer data between client and server

MVC – MODEL VIEW CONTROLLER:

➤ **MODEL:**

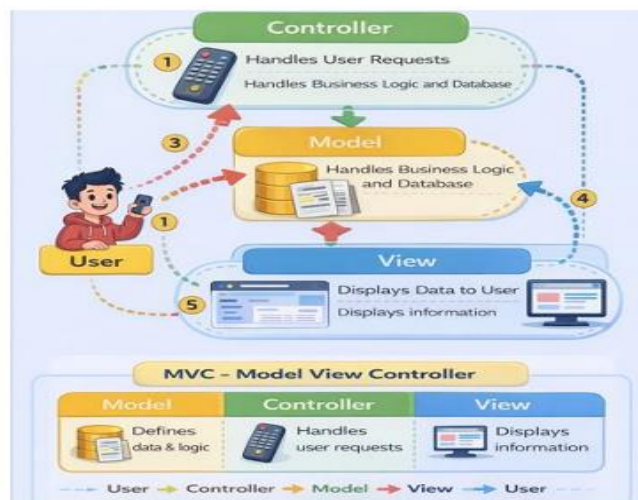
- Defines the data structure
- Handles business logic
- Represents application data
- (Example: Work done by motherboard)

➤ **VIEW:**

- Used to display data
- UI / Front-end
- (Example: TV screen)

➤ **CONTROLLER:**

- Contains control logic
- Handles user requests and responses
- Connects Model and View
- (Example: Remote control)



SPRING VS SPRING BOOT:

Feature	Spring	Spring Boot
Type	Framework	Framework built on Spring
Configuration	Manual	Auto-configuration
XML Required	Yes (mostly)	No
Server	External	Embedded
Setup Time	More	Very less
Boilerplate Code	High	Low
Production Ready	Needs setup	Yes (Actuator, metrics)
Dependency Management	Manual	Starter dependencies

MAVEN PROJECT:

- File → New → Maven Project
- Used to build our project using Maven.

WORK PROCESS:

- **pom.xml:**
 - pom file is used to run our file in a web server.
 - All XML details are present in the pom.xml file.
 - Maven is used to run Spring projects.
 - To add Spring, we need to add dependencies.
 - Dependencies can be added manually, but it is difficult
- **Spring Tools:**
 - To initialize a project, Spring Tools provide all dependencies.
 - All dependencies are given by the Spring Tool.
 - It is built on Eclipse, so it looks like Eclipse.
 - **Spring Tool Suite (STS):**
 - ✓ It will automatically handle all required things.